F78AB Actuarial and Financial Mathematics

Chapter 4: Nominal Interest Rates

4.1 Nominal Rates of Interest

Example 4.0

Say you put £100 into a savings account at a bank for a year. Which effective interest rate would you rather have

- 6% per annum?
- 2.9563% per half-year?

4.1 Nominal Rates of Interest

Throughout this section the time unit will be one year, unless stated otherwise.

- Consider a rate of interest i per annum.
  - **Effective interest** means that interest of £i is paid at the end of the year on £1 invested at the start of the year.
  - **Nominal interest** of $i^{(m)}$ per annum compounding m times a year means that £$i^{(m)}/m$ is paid at the end of each fraction $1/m$ of a year on £1 invested at the start of the year.

What will the timeline look like?

4.1 Nominal Rates of Interest

- The relationship between the effective interest rate i p.a. and the nominal rate of interest $i^{(m)}$ p.a. is

$$1+i=\left(1+\frac{i^{(m)}}{m}\right)^{m}$$

In example 4.0:

$1+i=\left(1+\frac{i^{(m)}}{m}\right)^{m}$

Example 4.1.1

The following are equivalent:

Effective Rate (p.a.)	Nominal Rate (p.a.)
$i = 0.03$	$i^{(2)} =$
$i = 0.06$	$i^{(12)} =$
$i =$	$i^{(4)} = 0.04$
$i =$	$i^{(3)} = 0.06$

$i^{(m)}$ per annum convertible m^{thly} or m times per year = effective rate of interest of $\frac{i^{(m)}}{m}$ per $\frac{1}{m}$ year

$1+i=\left(1+\frac{i^{(m)}}{m}\right)^{m}$

Exercise 4.1

The effective rate of interest is 3.5% per annum. Express this as the equivalent annual nominal rate of interest:

a) convertible quarterly. **0.034 550**

b) convertible 12 times a year. **0.034 451**

Exercise 4.2

The nominal rate of interest per annum convertible half-yearly is 5% per annum. Express this as the equivalent annual

a) effective rate of interest. **0.050 625**

b) nominal rate of interest convertible monthly. **0.049 487**

4.2 Nominal Rates of Discount

- Recall that $d = 1 - v$ is the **effective rate of discount per time unit**.
- d could be interpreted as the amount of interest payable at the start of the time unit that is equivalent to i payable at the end of the time unit.

Example 4.2.1

Generally, bank or savings accounts pay interest at the end of a set time period. However, Santander, a retail bank, launched a 3-year savings account in 2011 that paid interest upfront. No withdrawals were permitted during the 3 years.

https://www.theguardian.com/money/2011/nov/02/santander-savings-account-interest-upfront

Santander has launched a three-year fixed-rate bond that pays interest in advance, the first product of its kind.

The Upfront Interest bond, which is being marketed with a television advertising campaign fronted by US Open golf champion Rory McIlroy, is available to anyone with a Santander current account, or those savers who open one.



Blazing: Spanish owned bank Santander has launched a three year fixed bond which pays customers interest in advance

The bond can be opened in branches and by phone. The minimum deposit is £10,000 and capital cannot be accessed for three years.

But in a unique twist, interest is paid in full within six weeks of opening the bond. The rate is equivalent to 3.36 per cent gross per year.

For example, someone who deposits £10,000 this week will be paid just over £830 in net interest early next month.

The interest is paid into the Santander current account. No further interest is paid during the term and savers get their capital back in December 2014.

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4.2 Nominal Rates of Discount

- The *nominal discount rate* $d^{(m)}$ represents interest of $d^{(m)}/m$ paid at the start of each $1/m$ of a year.
- $d^{(m)}$ per annum convertible m^{thly} or m times per year is identical to an effective rate of discount $d^{(m)}/m$ per $1/m$ year.
- The relationship between the effective discount rate d p.a. and the nominal discount rate $d^{(m)}$ p.a. is $1-d=\left(1-\frac{d^{(m)}}{m}\right)^m$

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$1-d=\left(1-\frac{d^{(m)}}{m}\right)^m$

Example 4.2.2

The nominal discount rate convertible quarterly is 8% per annum.

Express this as the equivalent annual:

a) effective discount rate. **0.077 632**

b) effective interest rate. **0.084 166**

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$$1-d=\left(1-\frac{d^{(m)}}{m}\right)^{m}$$

Exercise 4.3

The effective rate of discount is 4% per annum. Express this as the equivalent annual nominal rate of discount:

a) convertible quarterly. **0.040 614**

b) convertible half-yearly. **0.040 408**

Exercise 4.4

The nominal rate of discount per annum convertible monthly is 6% per annum. Express this as the equivalent annual

a) effective rate of discount. **0.058 377**

b) nominal rate of discount convertible quarterly. **0.059 7005**

Equivalent Payments

	0	1/p	2/p	3/p	...	(p-1)/p	1
(1)	d						
(2)	$d^{(p)}/p$	$d^{(p)}/p$	$d^{(p)}/p$	$d^{(p)}/p$	...	$d^{(p)}/p$	
(3)		$i^{(p)}/p$	$i^{(p)}/p$	$i^{(p)}/p$	...	$i^{(p)}/p$	$i^{(p)}/p$
(4)							i

Example 4.2.3

£2500 is invested in a savings account for one year. The nominal rate of interest per annum convertible monthly for the first 3 months is 24% and the nominal rate of interest convertible quarterly for the next 9 months is 16% per annum. How much is in the account at the end of the year?

2984.29

Annuities payable *mthly*

- Assume m and n are positive integers.

Symbol	Meaning
	The value at time 0 of a level annuity payable *mthly* in arrears at the rate of 1 per unit time over the time interval $[0,n]$.
	The value at time 0 of a level annuity which pays $1/m$ in arrears m times per year for a period of n years.
	The value at time n of a level annuity payable *mthly* in arrear at the rate of 1 per unit time over the time interval $[0,n]$.

Useful Results

$$a_{\overline{n}|}^{(m)} = \sum_{t=1}^{nm} \frac{1}{m} v^{\frac{t}{m}} = \frac{1-v^n}{i^{(m)}}$$ T3Q1

$$a_{\overline{\infty}|}^{(m)} = \frac{1}{i^{(m)}}$$

$$\ddot{a}_{\overline{n}|}^{(m)} = \sum_{t=1}^{nm} \frac{1}{m} v^{\frac{t-1}{m}} = \frac{1-v^n}{d^{(m)}}$$ T3Q2

$$\ddot{a}_{\overline{\infty}|}^{(m)} = \frac{1}{d^{(m)}}$$

$$s_{\overline{n}|}^{(m)} = \sum_{t=1}^{nm} \frac{1}{m} (1+i)^{n-\frac{t}{m}} = \frac{(1+i)^n - 1}{i^{(m)}}$$ T3Q1

$$\ddot{s}_{\overline{n}|}^{(m)} = \sum_{t=1}^{nm} \frac{1}{m} (1+i)^{n-\frac{t-1}{m}} = \frac{(1+i)^n - 1}{d^{(m)}}$$ T3Q2

Exercise 4.5

Calculate the present value as at 1 January 2014 of the payments of $100 made on the first day of each quarter during the calendar years 2014 and 2015, if the effective rate of interest is 8% per annum.

$748.63

Exercise 4.6

An investor who pays income tax at 20% on all their income buys, at issue, a 25-year bond to be redeemed at par bearing semi-annual (twice per year) coupons of 8% per annum. What price will the investor pay per $100 nominal to obtain a redemption yield after income tax of 9% per annum effective?

$75.84

Requires knowledge of bonds (Chapter 11) so skip for now.

Annuities payable *mthly* where $m < 1$

$m < 1$: an annuity payable less frequently than annually.

For example $m = 0.5$ means payments are made once in 2 years i.e. biennially.

Exercise 4.7:

What is the value at time 0 of a series of 7 payments, each of amount 4, made at time 4, 8, 12, . . . , 28?

Example 4.2.4

- Suppose that k and r are integers greater than 1.
- Consider a series of payments, each of amount X, due at times $r, 2r, 3r, \ldots, kr$.
- What is the value of this series at time 0 on the basis of an effective interest rate i per unit time?

Example 4.2.4

The situation can be illustrated by replacing the payment of X due at time r by a series of r payments, each of amount Y, due at times $1,2,\ldots,r$, where Y is chosen such that the future value at time r of these r payments of Y is equivalent to the single payment they replace i.e. X at time r. This means that

$$Ys_{\overline{r}|} = X \quad \Rightarrow \quad Y = \frac{X}{s_{\overline{r}|}}$$

Hence the value of the annuity is: $\frac{X}{s_{\overline{r}|}} a_{\overline{kr}|}$

Exercise 4.8

A company expects to receive £10,000 from a customer. The amount will be split into ten equal instalments. The first instalment is due in one years' time and the remaining payments will be made at three year intervals thereafter. Calculate, to the nearest £, the present value of these future receipts at an effective rate of 7% per annum.

Answer: £4,419

Next Chapter

Chapter 5:
Real Interest Rates